

LAKSHAY BATRA

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EDUCATION

Thapar Institute of Engineering and Technology (TIET), Patiala

2023 – 2027

B.Tech in Computer Science & Engineering

Scholars Rosary Sr. Sec. School, Rohtak

CBSE (Class XII)

Graduated Mar 2023

Overall Percentage: 92%

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frameworks: React.js, Node.js, Express.js, Scikit-learn, XGBoost, Tailwind CSS

Databases: MongoDB, MySQL

Developer Tools: Git, GitHub, VS Code, Postman, REST APIs, JWT Auth

Core Concepts: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems

PROJECTS

IntelliFlow – Enterprise Workflow Automation

React.js, Node.js, Express.js, MongoDB

- Architected a request-to-execution automation system converting client requests into multi-sprint projects using a multi-factor matching algorithm (skills, workload, availability).
- Engineered **40+ RESTful API endpoints** with RBAC authorization, distributed approval workflows, lifecycle triggers, dependency management, and real-time cross-team aggregations.
- Built scalable frontend using **React.js and TypeScript** with modular UI components for responsive interface across 6 cross-functional teams.
- Implemented enterprise-grade security: **JWT authentication**, bcrypt (cost-12), OTP-based 2FA, CORS policies, NoSQL injection sanitization, XSS protection, and dual rate limiting.

Keystroke Authentication System

Python, Scikit-learn, XGBoost, SVM

- Developed a behavioral biometric security layer analyzing **31 unique timing features** (flight time, dwell time) to fingerprint users based on typing patterns.
- Benchmarked **XGBoost, SVM, and Random Forest** models, achieving a **94% accuracy rate** in distinguishing genuine users from impostors.
- Designed a robust data pipeline processing **200+ samples** per user, utilizing distinct training/testing splits to validate model generalization.
- Enhanced model discrimination capability through **feature standardization, hyperparameter tuning, and threshold optimization** to improve separation between **genuine** and **impostor** attempts.

POSITIONS OF RESPONSIBILITY

Entrepreneurship Development Cell (EDC), TIET

- Coordinated **logistics and operations** for flagship events like **E-Summit**, ensuring seamless execution for **500+ participants**.
- Facilitated **workshops** on **business modeling** and **pitching**, fostering a **startup culture** among the student body.

CERTIFICATIONS

Machine Learning Specialization – Stanford Online & DeepLearning.AI (Coursera)

2025

Full-Stack Web Development Bootcamp – Udemy (Dr. Angela Yu)

2025